

EasyWave: fast tsunami simulation tool for early warning

Andrey Babeyko (GFZ German Research Centre for Geosciences, Potsdam). Contact: babeyko@gfz-potsdam.de

Stand: Feb 2012

General description and motivation

EasyWave is a quick tool written in C++ to simulate tsunami generation and propagation in application to the early warning.

Specifics of the early warning, especially for local tsunamis, is that, on one hand, it requires a very prompt access to results of tsunami simulations, and, on the other hand, it is relatively tolerant to the simulation precision.

A prompt access to the simulation results is dictated by the time constraints available to the early warning procedure. For example, in GITEWS, it is 5-10 minutes after the earthquake. Usually, early warning systems store a large number of pre-computed tsunami scenarios in a databank of virtual tsunamis and then pick-up a scenario with the closest source model or build a composite scenario by combining individual pre-computed scenarios (the latter is valid in linear shallow water approximation only). Construction and maintenance of such a huge (usually, many terabytes) databank is itself a challenging IT- and management- problem. Moreover, this approach introduces an inevitable sampling error – in case of pre-computed full scenarios, the closest pre-computed source model does not necessarily coincide by location and magnitude with actual seismic observations. Not mentioning another assumptions about other fault parameters made for pre-computed scenarios.

An alternative way to get prompt simulation results is to run simulations in real- or supra-real time. That is what *EasyWave* does. The first hour of wave propagation on a 2-arc minute ETOPO-grid takes few seconds to be computed on a normal PC or notebook. That makes possible to follow in real time constantly improving source parameters. It is very important since the main uncertainty in early warning comes from the source. Especially, for local tsunamis.

Of course, this computation speed has to be paid for. *EasyWave* computes shallow water equations in linear approximation. That means it does not calculate coastal inundations and detailed runups directly. *EasyWave* brings the tsunami wave until the validity limit of the linear shallow water model (usually, 20-50 meter depth) and then estimates peak coastal tsunami amplitudes with Green's law (similar to the Japanese TEWS, see Kamigaichi, 2009) or runups with empirical relations like Chesley and Ward (2006). The code is open for other extrapolation schemes as well.

Author believes that this approach is absolutely valid for the early warning due to the following reasons.

- (1) Main uncertainties in tsunami simulations with runups come from:
 - a. source uncertainties
 - b. shallow bathymetry and coastal topography
 - c. numerical schemes and approximations itself.
- (2) In the case of early warning (some minutes after the earthquake), the main uncertainty comes from the source. Especially, the moment magnitude, which is a first-order tsunamigenic parameter, is subjected to changes and constant improvements.

- (3) To make realistic runup and inundation simulations, one must operate with very high-resolution shallow bathymetry and topography (10-50 m). Such datasets usually are available only for selected places like important coastal cities. If using coarse ETOPO or GEBCO grids, simulated runups are anyway only approximations – no matter which numerical schemes are being used.
- (4) Since there are so many uncertainties in real-time (first of all, uncertainties related to the source parameters), early warning operates not with meters and centimeters of runups but with warning levels like “no tsunami”, “minor tsunami” and “major tsunami”. That is why I’ve mentioned at the beginning that the early warning is “relatively tolerant to the simulation precision”.

Summarizing, *EasyWave* employs a “light-weighted” numerical scheme to simulate tsunami wave propagation and runups, which is however, absolutely reasonable for the goals of early warning.

Source Modeling

There are two possibilities to initiate tsunami propagation: (1) via earthquake fault(s) parameters and (2) via direct input of the initial wave profile. Both source options assume instant sea-surface displacement. Time-dependent forcing is presently not implemented.

1. Co-seismic displacement is modeled according to the classical Okada (1985) model for the uniform-slip rectangular rupture in a homogeneous halfspace. Any number of faults could be given in an input file which allows user to simulate ruptures with complicated non-uniform slip distributions.
There is no fixed set of input fault parameters in *EasyWave*. General rule – a fault should be sufficiently defined. For example, one can literally specify the whole set of Okada’s parameters: location, depth, length, width, slip, strike-, dip- and rake-angle, or give fault location, orientation and earthquake moment magnitude. In the latter case, scaling relations (presently, “JMA-scaling law” from Kamigaichi, 2009) would be implied to estimate rupture geometry and amount of slip.
2. Alternatively, initial wave profile could be directly given on the program input. In this case input 2D-map should be stored in text- or binary- Golden Software GRD-format.

Wave propagation

EasyWave computes wave propagation in the linear approximation of the long-wave theory in spherical coordinates. The applied numerical scheme follows the well-known TUNAMI-F1 algorithm [*TIME Project*, 1997], and is leap-frog explicit time-stepping on a staggered finite-difference grid. Boundary conditions presume full reflection along shorelines.

Solving the linear long wave equations is computationally very efficient, however, this approximation is no longer valid in coastal areas with water depths shallower than ~50 m [*Shuto*, 1991], where the nonlinear bottom friction and advection terms become non-negligible. This fact, combined with the rather coarse bathymetry resolution (e.g., 2 arc min corresponds to ~3.6 km grid step), did not allow us to calculate coastal run-ups directly. Instead, we accepted the approach employed and extensively verified by the Japanese tsunami early warning system [*Kamigaichi*, 2009]. In this approach, peak offshore tsunami amplitudes are first computed at some offshore positions (typically, in 50-100 m water depth) and then extrapolated into the peak coastal tsunami amplitudes (PCTA) using the Green’s law accounting for wave shoaling.

Kamigaichi [2009] found that thus derived PCTAs agree well with the means (and medians) of tsunami amplitudes computed directly at a coastline at a much finer mesh in each coastal subsection. In this respect, extrapolated PCTA's can be considered as representative estimates of the tsunami impact at the coast. As also noted by Kamigaichi [2009], this approach cannot discriminate between direct and reflected waves and, hence, may somehow overestimate PCTA's.

Program output

EasyWave produces both 1D- and 2D-outputs. Wave time series could be recorded at any given location (point-of-interest: POI) in a simple text-table format. Hazard relevant data such as estimated time of arrival (ETA) and estimated wave height (EWH) are provided as well. Additionally, 2D outputs of the wave propagation are stored at regular time intervals as binary GRD-files (Golden Software grid file format). These 2D-output is compact since only part of the grid corresponding to the actual propagation area is written. Another 2D-outputs (also in binary GRD-format) contain maximum wave height and tsunami arrival times.

Validation of the EasyWave

Publication submitted: Greenslade et al. "An Assessment of the Diversity in Scenario-based Tsunami Forecasts in the Indian Ocean".

Additionally *EasyWave* predictions have been compared to another two codes- TUNAMI-N2 and TsunAWI, for a scenario south of Java island. *illustrations to be prepared...*

References

- Chesley, S. R. and Ward, S. N., 2006, A quantitative assessment of the human and economic hazard from Impact-generated tsunami, *J. Nat.Hazard*, **38**, 355–374.
- Kamigaichi, O. (2009), Tsunami forecasting and warning. *Encyclopedia of Complexity and System Science*, pp. 9592-9617 , Springer.
- Okada, Y., 1985, Surface deformation due to shear and tensile faults in a half-space, *Bull. Seism. Soc. Am.* **75**, 1135-1154.
- Time project (1997), IUGG/IOC Time Project: numerical method of tsunami simulation with the leap-frog scheme. *IOC Manuals and Guides* 35, pp. 126.